



Reproductive toxicity of Roundup herbicide exposure in male albino rat

Folarin O. Owagboriaye^{a,*}, Gabriel A. Dedeke^b, Kehinde O. Ademolu^b, Olarenwaju O. Olujimi^c, Joseph S. Ashidi^d, Aladesida A. Adeyinka^b

^a Department of Zoology, Faculty of Science, Olabisi Onabanjo University Ago-Iwoye, Ogun State, Nigeria

^b Department of Pure and Applied Zoology, College of Bioscience, Federal University of Agriculture, Abeokuta, Ogun State, Nigeria

^c Department of Environmental Management and Toxicology, College of Environmental Management, Federal University of Agriculture, Abeokuta, Ogun State, Nigeria

^d Department of Plant Science, Faculty of Science, Olabisi Onabanjo University Ago-Iwoye, Ogun State, Nigeria

ARTICLE INFO

Keywords:

Roundup
Agriculture
Food poison
Herbicide
Reproduction

ABSTRACT

The incidence of infertility in human is on the increase and the use of Roundup herbicide and presence of its residues in foodstuff is a major concern. This study therefore aim to assess the effect of Roundup on the reproductive capacity of 32 adult male albino rats randomized into 4 groups of 8 rats per group orally exposed to Roundup at 3.6 mg/kg body weight(bw), 50.4 mg/kg bw and 248.4 mg/kgbw of glyphosate concentrations for 12 weeks while the control group was given distilled water. Serum level of reproductive hormone (testosterone, luteinizing hormone (LH), follicle stimulating hormone (FSH) and prolactin), oxidative stress indices in the testicular tissue, epididymal sperm morphology assessment and testicular histopathology of the rats were used as a diagnostic marker of reproductive dysfunction. Significant ($p < 0.05$) alterations in the level of all the reproductive hormones and oxidative stress markers assayed were observed in rats exposed to Roundup. Significant reductions ($p < 0.05$) in sperm count, percentage motility and significant ($p < 0.05$) increased in abnormal sperm cells were observed in the exposed rats. Histopathologically, severe degenerative testicular architectural lesions were seen in the Roundup exposed rats. Roundup may interfere with spermatogenesis and impair fertility in male gonad.

1. Introduction

Agriculture belongs to the real sector of the Nigerian economy. The roles of agriculture in the Nigerian economy remain significant despite the over-dependence on the oil sector. The increase in the human population has made Nigerian farmers intensify agricultural production with consequent high incidences of pest, weed and plant diseases. Farmers are responding to these problems through mechanization and the use of agrochemicals (e.g. herbicides to control weeds and pesticides to minimize losses to pests). One of the commonly used herbicide worldwide is glyphosate-based herbicides such as Monsanto Roundup (Mesnage et al., 2015).

Glyphosate, an N-(phosphonomethyl) glycine, is the active ingredient in Monsanto Roundup herbicide. It is a broad spectrum herbicide widely used to eliminate unwanted plants both on agricultural and non-agricultural landscapes (Temple and Smith, 1992). It has been reported that glyphosate is the top ranked herbicide in the UK arable crop production (Garthwaite, 2010). Also, Germany use of glyphosate has been estimated to cover 4.3 million hectares (39%) of agricultural land each year, with nearly two-thirds applied to winter wheat and winter

barley (Steinmann et al., 2012). Similarly, 50% to 60% of sunflower crops in France, Romania and Hungary are treated before harvest with glyphosate and it is the most commonly used herbicide in Africa (Gianessi and Williams, 2011). Increased yield of agricultural productions up to 55% as a result of increase use of the Roundup in Nigeria has been reported by Chikoye et al. (2007). Glyphosate is also used as a desiccant on cereals, oilseed rape and maize (Monsanto, 2010). As a result, residues of this herbicide has been reportedly detected in foodstuffs and also in drinking water contaminated via rain, surface runoff and leaching into groundwater, thereby increasing likely routes of exposure in animals and humans (Bohn et al., 2013; EFSA, 2014).

Glyphosate's primary mode of action is to block an enzyme 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) of shikimate pathway present in plants and some bacteria which are responsible for the biosynthesis of chorismate, an intermediate in the biosynthesis of phenylalanine, tyrosine and tryptophan essential aromatic amino acid. This pathway is known to be absent in animals, thus, generally believed that glyphosate poses no effects in animals making this mechanism of action exclusive to plants (Cerdeira et al., 2007).

Unfortunately, irreversible liver damage on rats consuming

* Corresponding author.

E-mail address: fobillion@yahoo.com (F.O. Owagboriaye).

4.87 mg/kg body weight (bw), of glyphosate every 2 days over 75 days has been reported (Benedetti et al., 2004). Haematological, biochemical and oxidative alterations had also been observed in mice exposed to 50 and 500 mg/kg bw/day of glyphosate (Jasper et al., 2012). A report by the European Food Safety Authority (EFSA, 2016) revealed that Glyphosate is unlikely to be genotoxic, neurotoxic or toxic for the reproduction or development and is unlikely to pose a carcinogenic hazard to humans at acceptable daily intake (ADI) of 0.5 mg/kg bw/day. However, Mesnage et al. (2015) and Mesnage et al. (2017) had shown the transcriptomic and multiomic profile analysis of liver and kidney damage in rats exposed to a chronic ultra-low dose of 0.1 ppb Roundup equivalent to 4 mg/kg bw of glyphosate consumed which are well below the ADI value. Mesnage et al. (2015) also reported that the observed alterations in gene expression were consistent with fibrosis, necrosis, phospholipidosis, mitochondrial membrane dysfunction and ischemia, which correlate with and thus confirm observations of pathology made at an anatomical, histological and biochemical level. Teratogenic effect of glyphosate-based herbicide has been documented in mammals (Antonioni et al., 2012). It has also been linked to the production of reactive oxygen species (ROS) resulting in oxidative stress in animals (De Liz et al., 2013).

Reports have also linked the increasing trend of male reproductive abnormality recorded in some countries with possible exposures to endocrine disrupting chemicals that could interfere with endocrine homeostasis (Bergström et al., 1996; Möller, 2001). Certain male reproductive damages such as reduced sperm count, testicular lesions, sperm motility changes and sperm morphologic abnormality had been observed in animals exposed to some pyrethroid pesticides (Xia et al., 2004; Song et al., 2008). However, a significant reduction in serum testosterone concentrations and changes in the testicular morphology of male Wistar rats treated with Roundup at 5 mg/kg bw, 50 mg/kg bw and 250 mg/kg bw of glyphosate have been documented (Romano et al., 2010). On the contrary, the German Federal Institute for Occupational Safety and Health (2016) recently evaluated three generational studies by Schroeder and Hogan (1981) and Bhide (1988) where no adverse reproductive effect in rats exposed to glyphosate at top dose of 15 mg/kg bw/day was observed. In addition, no evidence of reproductive toxicity was observed in rats exposed to glyphosate at an exaggerated dose of 2000 mg/kg bw/day (Takahashi, 1997). Likewise, Dhinsa et al. (2007) reported a decrease in homogenization resistant spermatids in the caudal epididymides with no impact on fertility or reproductive success after administration of glyphosate at 15000 ppm. It is on these bases that the German Federal Institute for Occupational Safety and Health (2016) summarized in its report that very large database submitted by different applicants and from published scientific literature to evaluate the reproductive and developmental toxicity of glyphosate in rat did not provide evidence of reproductive toxicity or teratogenicity. The institute also claimed that the few observed effects were small, of equivocal relevance and confined to parentally toxic dose levels. The German Federal Institute for Occupational Safety and Health (2016) also reported that no convincing evidence of reproductive or developmental effects of glyphosate may be derived from epidemiological studies or from *in vitro* or *in vivo* studies on different aspects of reproduction and therefore, proposed no classification and labeling of glyphosate for reproductive or developmental effects. In an effort to provide insight on the reproductive toxicity of glyphosate, this study was therefore designed to assess reproductive hormones, epididymal sperm morphology, sperm count and motility, testicular antioxidant defense system and architectural structure of testes of albino rats exposed to Monsanto Roundup.

2. Materials and methods

2.1. Chemicals

The herbicide product tested was Roundup (Monsanto Europe S.A./

N.V., Antwerp, Belgium, O611, F- 1059 3379) in a commercial formulation (360 g/L of glyphosate in the form of 441 g/L potassium salt). It was stored in its original container and kept in a cool, dry and well-ventilated area at room temperature throughout the experiment.

2.2. Experimental animals

Thirty-two apparently healthy adult male albino rats (200 ± 20 g) obtained from the breeding section of the animal house of the Department of Zoology, Olabisi Onabanjo University Ago-Iwoye Nigeria, were used for this study. The rats were initially acclimatized under the laboratory conditions of $25 \pm 5^\circ\text{C}$ and $65 \pm 5\%$ Relative Humidity in a well-ventilated experimental animal house for one week before the commencement of the study. The rats were allotted into individual wooden cages ($65\text{ cm} \times 35\text{ cm} \times 50\text{ cm}$) and were allowed free access to standard laboratory rat chow and water. The experimental protocol was conducted in accordance with the regulations of the local ethics committee in animal care unit of Olabisi Onabanjo University, Ago-Iwoye, Ogun State, Nigeria. The animal experiment was performed according to ethical guidelines of animal experimentation (regulation CEE 86/609).

2.3. Experimental design

The rats were randomized into four experimental groups consisting of eight rats per group. Three groups were administered by oral gavage with 0.01 mL/kg bw/day, 0.14 mL/kg bw/day and 0.69 mL/kg bw/day of Monsanto Roundup herbicide corresponding to 3.6 mg/kg bw/day, 50.4 mg/kg bw/day and 248.4 mg/kg bw/day of glyphosate consumed respectively, while rats in control group were administered with distilled water for a period of 12 weeks. Roundup was diluted in a watery suspension and administered, with gavage, to the rats once a day in a volume of 0.25 mL/100 g of body weight as earlier described by Romano et al. (2010). The dose regime was based on no observed adverse effect level (NOAEL) for developmental toxicity in rats as reported (Williams et al., 2000, 2012; Benedetti et al., 2004). The values are within the limits of NOAEL and equivalent to 1/1000, 1/100, and 1/20 of LD50 in rats as described in Larini (1999). At the end of the exposure period, blood sample, epididymal sperm cells and testicular tissue were collected from the rats in each group for laboratory analysis.

2.4. Sample collections

Blood samples were collected into clean sample tubes by retro-orbital sinus with micro hematocrit tube in baseline conditions between 8:00 A.M. and 10:00 A.M. The blood samples were centrifuged at $2500 \times g$ for 10 min and the obtained sera samples were stored in ice cold until used for the hormonal assay. The rats were sacrificed by cervical dislocation and testes were removed by dissection and washed using saline solution. Parts of this tissue were used to assay for antioxidant while the remaining parts were subjected to histopathological examination. Epididymis of the rats was dissected out and sperm cells were collected into 10 mL of 0.87% warmed normal saline for sperm abnormality assessment.

2.5. Hormonal assay

The sera obtained were analyzed to determine the concentration of testosterone, follicle stimulating hormone (FSH), prolactin and luteinizing hormone (LH) using the Enzyme-Linked immunosorbent Assay (ELISA). The ELISA kits were obtained from Biocheck, USA.

2.6. Antioxidant parameters analysis in testes tissue

Lipid peroxidation end product, malondialdehyde (MDA), was measured as thiobarbituric acid reactive substance (TBARS) as de-

scribed in Okhawa et al. (1979). The levels of reduced glutathione (GSH) were assayed according to Hassan and Barakat (2008). The activities of antioxidant enzymes were conducted according to standard procedure; catalase (CAT) activity was according to Johansson and Borg (1979), superoxide dismutase (SOD) was according to Soon and Tan (2002) and the method of Rotruck et al. (1973) was employed for the assay of glutathione peroxidase (GPx).

2.7. Epididymis sperm morphology assay and assessment

Sperm motility and count were carried out using an improved method of Hassan and Barakat (2008). Sperm morphology assay was according to Bakare et al. (2005). Epididymal sperm cells were suspended in normal saline and 1% Eosin Y stain. The smear was prepared and the slides were allowed to air-dried and coded for subsequent microscopic examination at $\times 1000$ magnification. 1000 sperm cells were assessed for morphological abnormalities for each rat.

2.8. Histopathological examination of the testes

A portion of the testes excised from rats in each group was fixed in Bouin's fluid for 48 h and routinely processed for paraffin embedding. Sections of 4 μm thickness were taken serially with a Rotary Microtome and then, processed in alcohol-xylene series and were stained with hematoxylin and eosin (H&E). The prepared slides were examined at $\times 400$ magnification.

2.9. Data analysis

Data obtained were presented as Mean $\pm$ Standard Error of Mean (SEM). Statistical analyses for all measurements were performed using Statistical Package for Social Sciences (SPSS) version 20.0. The mean, standard error of the mean, Analysis of Variance (ANOVA) and Correlation coefficient were conducted. Post Hoc test was done using the Student-Newman-Keuls (SNK). $P < 0.05$ was considered to be statistically significant.

3. Results

3.1. Level of reproductive hormones in the blood

Mean level of reproductive hormone (testosterone, FSH, LH and prolactin) measured in the blood of the rats exposed to Roundup at different concentrations of glyphosate after 12 weeks is presented in Table 1. Testosterone, FSH and LH levels were statistically significant ($p < 0.05$) in the Roundup exposed rats compared to the control. On the other hand, the level of prolactin hormone in the blood of the rats was an inverse of testosterone, FSH and LH. Prolactin was observed to be significantly higher ($p < 0.05$) in Roundup exposed rats. The observed increase or decrease in the hormonal level appeared to be dose dependent.

Table 1

Level of the reproductive hormone in the blood of the rats exposed to Roundup at different concentrations of glyphosate.

Treatments	Testosterone (ng/ml)	FSH (mIU/ml)	LH (mIU/ml)	Prolactin (ng/ml)
Control	3.17 $\pm$ 0.12 ^a	9.49 $\pm$ 0.26 ^a	8.29 $\pm$ 0.34 ^a	2.36 $\pm$ 0.21 ^d
3.6 mg/kg bw	2.75 $\pm$ 0.09 ^b	7.84 $\pm$ 0.26 ^b	7.11 $\pm$ 0.09 ^b	3.09 $\pm$ 0.11 ^c
50.4 mg/kg bw	2.10 $\pm$ 0.07 ^c	6.11 $\pm$ 0.33 ^c	5.42 $\pm$ 0.35 ^c	4.05 $\pm$ 0.19 ^b
248.4 mg/kg bw	1.07 $\pm$ 0.07 ^d	3.19 $\pm$ 0.12 ^d	2.18 $\pm$ 0.12 ^d	6.54 $\pm$ 0.35 ^a

^{a,b,c,d}Mean ($\pm$ Standard deviation) in the same column having similar superscript are not significantly different at $P < 0.05$.

3.2. Level of antioxidant parameters in the testes tissue

Table 2 shows the level of oxidative stress markers in testes of the rats exposed to Roundup at different concentrations of glyphosate. Mean levels of SOD, GSH, CAT and GPx in the testes of the Roundup exposed rats were significantly ($p < 0.05$) reduced compared to the control rats and the reduction was observed to be dose dependent. However, the mean level of MDA in the testes was statistically significant ($p < 0.05$) in the control rats and rises with increase concentrations of the glyphosate exposure.

3.3. Sperm motility and count

The percentage sperm motility and sperm count observed in the rats exposed to Roundup at varying concentrations of glyphosate were statistically significant ($p < 0.05$) compared to the control. The lowest sperm count was observed in the rats exposed to Roundup at 248.4 mg/kg bw of glyphosate (Fig. 1). However, lowest percentage sperm motility was observed in the rats exposed to Roundup at 50.4 mg/kg bw of glyphosate.

3.4. Sperm morphology assay

Sperm abnormalities induced in rats exposed to Roundup at different concentrations of glyphosate are presented in Fig. 2 and Table 3. Total abnormal sperm cells of 111.00 $\pm$ 4.76, 188.75 $\pm$ 4.11 and 238.50 $\pm$ 1.73 in rats exposed to Roundup at 3.6, 50.4 and 248.4 mg/kg bw of glyphosate respectively were observed. These values are fractions of the 1000 sperm cells assessed and were statistically significant ($p < 0.05$) compared to the control (19.50 $\pm$ 2.52). Abnormal sperm cells with a bent tail, without a head, damaged head, without a hook, banana shape and without tail were observed in the rats exposed to Roundup (Fig. 2) and these values were statistically significant ($p < 0.05$) with increasing concentrations of glyphosate (Table 3).

3.5. Histopathological assessment of the testes

The photomicrographs of the histopathology of testes of rats exposed to Roundup at different concentrations of glyphosate and the control rats are displayed in Fig. 3. Control group revealed the normal cellular architectural structure of the tissues while progressive degenerative lesions were observed in rats testes exposed to different concentrations of glyphosate (Fig. 3). Degeneration of testicular tissue, tubular hyalinization, a moderately depleted amount in the generation of sperm cells and interstitial cells were observed in the testes of the rats exposed to Roundup at 3.6 mg/kg bw of glyphosate. Severe testicular distortions characterized by vacuolation of the seminiferous tubule, alterations in the generation of sperm cells and reductions of interstitial cells in rats exposed to Roundup at 50.4 mg/kg bw of glyphosate were observed. Rats exposed to Roundup at 248.4 mg/kg bw of glyphosate showed marked vacuolation of the seminiferous tubule, loss of interstitial cells with marked reductions in sperm cells generation.

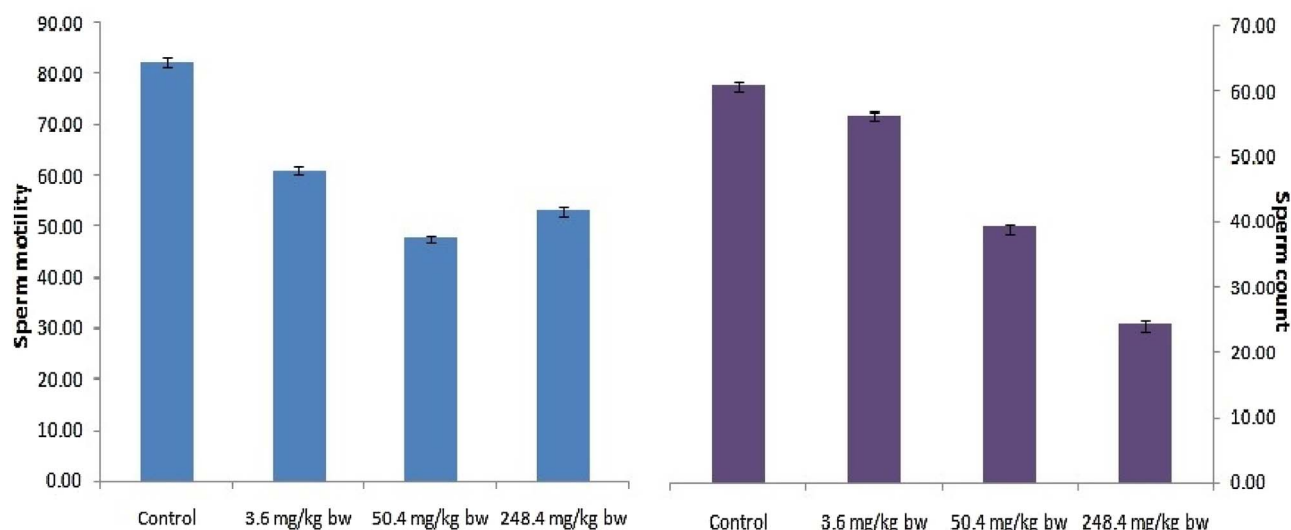
3.6. Relationship of reproductive hormones in blood and the testicular antioxidant defense system with percentage sperm motility, sperm count and total sperm abnormality count

The results of the relationship of reproductive hormones in blood and the testicular antioxidant defense system with percentage sperm motility, sperm count and total sperm abnormality count in the rats are presented in Table 4. The relationship between blood levels of reproductive hormones (except prolactin) and testes antioxidant (except MDA) with sperm motility and sperm count were positive and significantly strong ($p < 0.01$). However, the blood level of prolactin and testes level of MDA were negatively correlated with sperm motility

Table 2

Level of oxidative stress markers in the testes of rats exposed to Roundup at varying concentrations of glyphosate.

Treatments	SOD (U/mg protein)	GSH (U/g tissue)	CAT (U/mg protein)	GPx (U/mg protein)	MDA (nmol/g tissue)
Control	6.88 ± 0.26 ^a	11.96 ± 0.16 ^a	10.39 ± 0.41 ^a	17.51 ± 0.42 ^a	21.97 ± 0.41 ^d
3.6 mg/kg bw	5.07 ± 0.03 ^b	9.04 ± 0.11 ^b	8.93 ± 0.18 ^b	14.11 ± 0.30 ^b	27.92 ± 0.18 ^c
50.4 mg/kg bw	3.65 ± 0.15 ^c	5.15 ± 0.06 ^c	6.21 ± 0.16 ^c	9.27 ± 0.28 ^c	38.90 ± 0.23 ^b
248.4 mg/kg bw	0.93 ± 0.46 ^d	2.37 ± 0.14 ^d	3.40 ± 0.21 ^d	5.10 ± 0.51 ^d	49.02 ± 0.32 ^a

^{abcd}Mean (± Standard deviation) in the same column having similar superscript are not significantly different at P < 0.05.**Fig. 1.** Sperm motility (%) and sperm count (× 10⁶ cells/ml) in rats exposed to Roundup at different concentrations of glyphosate.

and sperm count ($p < 0.01$). On the other hand, the relationship between blood levels of reproductive hormones (except prolactin) and testes antioxidant (except MDA) with total sperm abnormality count were strong and significantly negative ($p < 0.01$).

4. Discussion

Reduction in testosterone level observed in the exposed rats could be added to the negative impact of glyphosate in the Roundup consumed on the architectural structure of their testes and consequently affect the number and function of its Leydig or interstitial cells since these cells are the site of testosterone production and secretion (Zitzmann, 2008). Progressive decrease in the serum FSH and LH of the rats exposed to Roundup agree with Romano et al. (2010) and Nithya and Elango (2015) who also observed reductions in the level of male reproductive hormones in an animal exposed to Roundup and lindane pesticides respectively. The action underlying these reductions may be attributed to alteration in gonadotropin-releasing hormone (GnRH) in the hypothalamus. But the analysis on GnRH was not carried out in this study to ascertain this claim. Therefore, more researches on Roundup exposure in this area are suggested. Of great interest is the high level of serum prolactin (hyperprolactinemia) observed in the rats exposed to Roundup in this study. Horseman and Gregerson (2006) submitted that Hyperprolactinemia impairs male gonadal functions by acting at various levels, possibly by decreasing gonadotropin-releasing hormone (GnRH) pulse generator activity and might be inhibiting testosterone secretion at the level of Leydig cells as well. However, high level of prolactin observed in the rats exposed to Roundup could also be responsible for the corresponding decrease in their serum testosterone level.

Increased level of MDA and reduced activities of the antioxidant enzymes in the testes of the exposed rats may suggest induction of oxidative stress by the Roundup. It is worthy to note that glyphosate

would have induced production of ROS in the testes of the exposed rats, consequently induced the peroxidation of polyunsaturated fatty acids in the membrane of the testes which led to the formation of MDA, one of the by-products of lipid peroxidation as previously observed in Jasper et al. (2012). However, reductions in the activities of GSH, SOD and CAT in the testes of the exposed rats may be attributed to probably the excessive production of free radicals or ROS the glyphosate may have inflicted on the testes after the exposure. GPx uses GSH for its enzymatic activities. It also plays a major role in detoxification, GSH maintenance and homeostasis. Thus, reduction in the activities of GPx in the testes of the exposed rats could be a result of the observed reduction in their testicular GSH. However, it is reasonable to relate the presence of lipid peroxidation and reduced GSH observed in the testes of the exposed rats in this study to the reduction of their SOD, CAT and GPx activities which in turn, can be added to the oxidative stress inflicted on the testes tissue by the Roundup.

Reductions in epididymal sperm count and percentage sperm motility observed in rats exposed to Roundup may be attributed to induction of ROS in the testes and epididymis of the exposed rats. Shama et al. (2013) observed reduced sperm count and motility in rats exposed to cypermethrin due to ROS induced by the pesticide in the testes of the exposed rats. Although it has been reported in El-Tohamy and El-Nattat (2010) that low level of ROS is required for normal sperm function while the high level of ROS clearly impaired fertility. El-Tohamy and El-Nattat (2010) also submitted that lipid peroxidation is considered to be the key mechanism of ROS-induced sperm damage, which led to a loss of sperm motility. As reported by Mylchreest et al. (2002), a decrease in testosterone level is associated with marked reduction in epididymal sperm number. The finding of this study confirms these reports. Hyperprolactinemia in men has been shown to typically resulted into hypogonadism which led to decreased libido, erectile dysfunction and abnormal semen quality (Corseello et al., 2003), in a similar trend, Prolactin receptors have been demonstrated in the

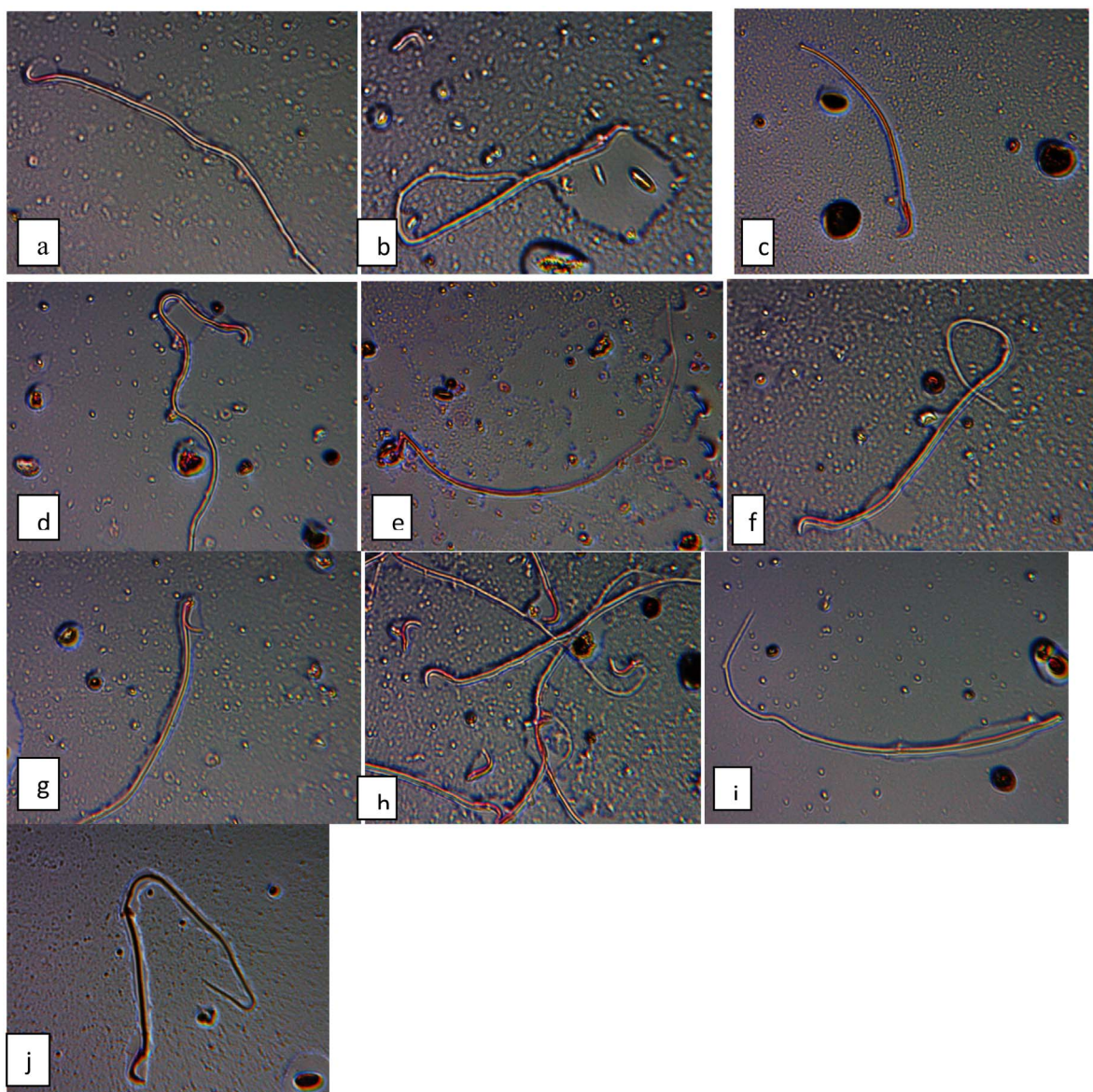


Fig. 2. Abnormal sperm cells induced in rats exposed to Roundup at varying concentrations of glyphosate (a) normal sperm cell (b) sperm without hook and bent tail (c) without tail (d) banana shape (e) sperm with damaged head (f) bent tail (g) damaged hook (h) sperm with several broken hook (i) without head (j) bent body and tail.

spermatogonia, seminiferous epithelium, and spermatocytes, suggesting that prolactin might function in normal spermatogenesis (Hondo et al., 1995). Hence, low sperm count and motility observed in this study among the exposed rats to Roundup may also be attributed to the increased in their serum prolactin.

Spermatogonia effect of Roundup is revealed in this study. Bakare et al. (2016) have stated that exact mechanisms for the increase in the frequency of abnormal sperm caused by chemical pollutants are not

clear. However, induction of abnormal sperm cells could be as a result of an abnormal chromosome (Bruce et al., 1974), minor alterations in testicular DNA (Giri et al., 2002) or point mutation (Narayana et al., 2002). The occurrence of sperm head abnormalities has been attributed to the chromosomal aberrations that occur during the packaging of genetic material in the sperm head or occurrence of a point mutation in testicular DNA (Bruce and Heddl, 1979). Thus, glyphosate interactions with the genetic material in the testes of the exposed rats during

Table 3
Sperm morphological abnormality count in rats exposed to Roundup at different concentrations of glyphosate.

Concentration	Bent tail	Without Head	Damaged Head	Without Hook	Banana Shape	Without Tail	Total Abnormal	Percentage Abnormalities
Control	11.50 ± 1.29 ^d	0.00 ± 0.00 ^d	1.25 ± 1.26 ^d	1.00 ± 0.82 ^d	5.75 ± 0.96 ^d	0.00 ± 0.00 ^c	19.50 ± 2.52 ^d	1.95 ± 0.25 ^d
3.6 mg/kg bw	28.75 ± 0.96 ^c	11.25 ± 1.26 ^c	21.75 ± 1.71 ^c	8.75 ± 0.96 ^c	30.25 ± 1.26 ^c	10.25 ± 1.26 ^b	111.00 ± 4.76 ^c	11.10 ± 0.48 ^c
50.4 mg/kg bw	37.25 ± 1.26 ^a	17.75 ± 1.26 ^b	36.75 ± 1.89 ^b	20.50 ± 1.29 ^b	58.50 ± 2.65 ^a	18.00 ± 3.56 ^a	188.75 ± 4.11 ^b	18.88 ± 0.41 ^b
248.4 mg/kg bw	32.00 ± 0.82 ^b	35.75 ± 1.26 ^a	59.50 ± 2.89 ^a	48.00 ± 2.58 ^a	42.25 ± 1.26 ^b	21.00 ± 3.37 ^a	238.50 ± 1.73 ^a	23.85 ± 0.17 ^a

^{abcd}Mean (± Standard deviation) in the same column having similar superscript are not significantly different at $P < 0.05$. Mean (± Standard deviation) are fractions of the 1000 sperm cells assessed

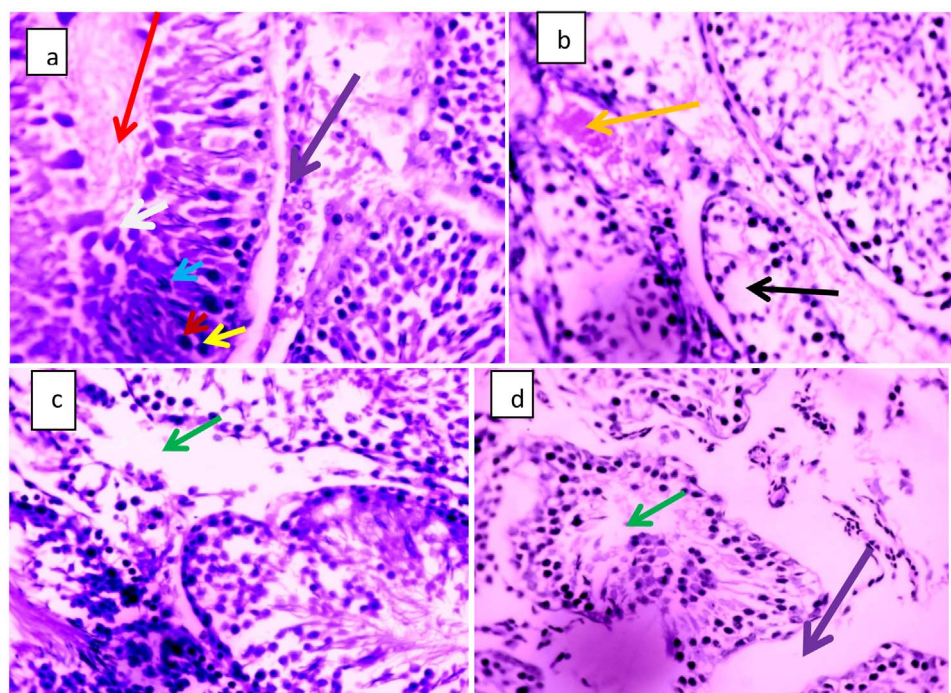


Fig. 3. Testes tissue (H & E stain; $\times 400$) of (a) Control showing normal seminiferous tubule (red arrow), interstitial cells (purple arrow), spermatogonia (yellow arrow), primary spermatocytes (dark red arrow), secondary spermatocytes (blue arrow) and spermatids (white arrow) (b) Rats exposed to Roundup at 3.6 mg/kg bw of glyphosate showing degeneration of testicular tissue (black arrow) and tubular hyalinization (orange arrow) with moderately depleted amount in all the stages of sperm cells (c) Rats exposed to Roundup at 50.4 mg/kg bw of glyphosate showing marked vacuolation of the seminiferous tubule (green arrow) affecting all the generations of sperm cells and moderate reduction of and scattered interstitial cells (d) Rats exposed to Roundup at 248.4 mg/kg bw of glyphosate with marked testicular lesions; vacuolation of the seminiferous tubule, loss of interstitial cells with marked reductions in all the generations of sperm cells. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Relationship of reproductive hormones in blood and the testicular antioxidant defense system with percentage sperm motility, sperm count and total sperm abnormality count.

		Sperm Motility	Sperm count	Total sperm abnormality
Blood	Testosterone	0.654**	0.841**	−0.801**
	FSH	0.738**	0.865**	−0.857**
	LH	0.668**	0.815**	−0.803**
	Prolactin	−0.638**	−0.844**	0.806**
Testes	SOD	0.749**	0.847**	−0.857**
	GSH	0.762**	0.892**	−0.889**
	CAT	0.701**	0.884**	−0.858**
	GPx	0.737**	0.899**	−0.878**
	MDA	−0.708**	−0.889**	0.861**

* Correlation is significant at $P < 0.01$ (Pearson correlation).

spermatogenesis which can probably lead to the observed increase in their abnormal sperm cells are reasonably suggested. [Kamijima et al. \(2004\)](#) also observed similar increased in percentage of abnormal sperm cells among insecticide sprayers exposed to Pyrethroids. The observed increased in abnormal sperm cells may also be attributed to the possible oxidative stress inflicted on the testes of the exposed rats as shown in the positive and significant relationship between the sperm abnormality count and oxidative stress indices analyzed. This observation is confirmed in [Nithya and Elango \(2015\)](#) who stated that oxidative stress contributes to defective spermatogenesis by reducing the antioxidant level in blood and testes thereby leading to male infertility. However, our finding is in contrast to [Akcha et al. \(2012\)](#) who reported no effects on the spermatozoa of oyster exposed to glyphosate.

The degenerative changes observed in the seminiferous tubule and interstitial cells of the rat testes are evidence of the toxicity of Roundup to the male reproductive system of the rat. As reported in [Thakur et al. \(2014\)](#), degenerative changes in the seminiferous tubule caused by a chemical agent may indicate the interference of the chemical with spermatogenesis process. Thus, glyphosate may have directly interfered

with the process of spermatogenesis in the testes of the exposed rats. This could probably be a pre-requisite for the observed abnormal sperm morphology produced by the damaged testes. This result is consistent with [Romano et al. \(2010\)](#) who had previously observed reduction in epithelium depth and larger diameter from the lumen in seminiferous tubule of rats exposed to Roundup.

It is important to state here that commercial formulation of Roundup is a mixture of glyphosate (active ingredient) with varying and unspecified surfactants, which are required for rapid penetration and absorption of the herbicide into the plant. [Jasper et al. \(2012\)](#) reported that the mechanisms of toxicity of glyphosate formulations are complicated. In a similar trend, [Bradberry et al. \(2004\)](#) stated that surfactants probably contribute to the acute toxicity of glyphosate formulations, but cannot potentiate its toxicity. This is an indication that the observed toxicity in this study may not directly result from the glyphosate composition but by the herbicide's complex and variable mixtures. Thus, our findings are not directly comparable to others testing glyphosate alone.

Nevertheless, the reproductive toxicity observed among the rats exposed to Roundup in this study may have been significantly influenced by the glyphosate. This cannot be ruled out, as glyphosate is known to be the active ingredient in commercial formulations of Roundup. Our finding is comparable and agrees with [Dallegrave et al. \(2007\)](#) who also observed reproductive dysfunctions in Wistar rat exposed to Roundup. However, the report by the [German Federal Institute for Occupational Safety and Health \(2016\)](#) claims that the study by [Dallegrave et al. \(2007\)](#) was performed with a commercial formulation and, thus, is also not useful for classification and labeling of the active substance. Therefore, additional studies testing the toxicity of glyphosate alone and comparing with commercial formulations are recommended in order to ascertain the reproductive toxicity status of glyphosate.

4.1. Conclusion

Our study has shown that Roundup has the capacity to induce reproductive toxicity in the male reproductive system of the exposed animal. It is also a potent endocrine disruptor. We can conclude that the disruption in the normal testicular cellular architecture observed in the rats exposed to Roundup in this study, which could probably, led to abnormal hormonal secretion and abnormal sperm properties may be attributed to the oxidative stress inflicted on the gonad of the exposed rats by the active ingredient in Roundup. This assumes significance and a public health concern considering increasing use of Roundup and presence of its residues in food and drinking water, thereby, increasing possible routes of exposure in human. Therefore, there is the need for a review on the use of Roundup in Nigeria.

Conflict of interest

All authors declare no competing interest in this study

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Authors roles

FOO conceived the experiment. FOO, GAD, KOA and OOO designed the experiments. FOO and JSA performed the experiment. FOO and AAA analyzed and interpreted the data. FOO drafted the manuscript which was critically revised by GAD, KOA, OOO and AAA and gave final approval. All authors agree to be accountable for integrity and accuracy in all aspects of the work.

Acknowledgements

We sincerely appreciate the permission of Department of Zoology, Faculty of Science, Olabisi Onabanjo University to use its facilities. We also thank Prof. A.A. Bakare of Department of Zoology, University of Ibadan Nigeria, for his generous donation of the Eosin Y stain used in this study.

References

- Akcha, F., Spagnol, C., Rouxel, J., 2012. Genotoxicity of diuron and glyphosate in oyster spermatozoa and embryos. *Aquat. Toxicol.* 107, 104–113.
- Antoniou, M., Habib, M.E.M., Howard, C.V., Jennings, R.C., Leifert, C., Nodari, R.O., 2012. Teratogenic effects of glyphosate-based herbicides: divergence of regulatory decisions from scientific evidence. *J. Environ. Anal. Toxicol.* S4 (S4). <http://dx.doi.org/10.4172/2161-0525.s4-006>.
- Bakare, A.A., Mosuro, A.A., Osibanjo, O., 2005. An in vivo evaluation of induction of abnormal sperm morphology in mice by land fill leachates. *Mutat. Res.* 582, 28–34.
- Bakare, A.A., Udoakang, A.J., Anifowoshe, A.T., Fadoju, O.M., Ogunsuyi, O.I., et al., 2016. Genotoxicity of titanium dioxide nanoparticles using the mouse bone marrow micronucleus and sperm morphology assays. *J. Pollut. Eff. Control* 4, 156.
- Benedetti, A.L., Vituri Cde, L., Trentin, A.G., Domingues, M.A., Alvarez-Silva, M., 2004. The effects of sub-chronic exposure of Wistar rats to the herbicide Glyphosate-Biocarb. *Toxicol. Lett.* 153, 227–232.
- Bergström, R., Adami, H.O., Möhner, M., Zatonski, W., Storm, H., Ekbohm, A., 1996. Increase in testicular cancer incidence in six European countries: a birth cohort phenomenon. *J. Natl. Cancer Inst.* 88, 727–733.
- Bhide, N., 1988. Report on Effect of Pesticides on Reproductive Process – Segment IV – Three Generation Reproduction Study with Albino Rats Using Glyphosate Technical of Excel Industries Ltd., Bombay. BVL-2328485, TOX955.
- Bohn, T., Cuhra, M., Traavik, T., Sanden, M., Fagan, J., Primicerio, R., 2013. Compositional differences in soybeans on the market: glyphosate accumulates in Roundup Ready GM soybeans. *Food Chem.* 153, 207–215.
- Bradberry, S.E., Proudfoot, A.T., Vale, J.A., 2004. Glyphosate poisoning. *Toxicol. Rev.* 23 (3), 159–167.
- Bruce, W., Heddle, J., 1979. The mutagenicity of 61 agents as determined by the micronucleus, Salmonella and sperm abnormality assays. *Can. J. Cytol. Genet.* 21, 319–334.
- Bruce, W.R., Furrer, R., Wyrobek, A.J., 1974. Abnormalities in the shape of murine sperm after acute testicular x-irradiation. *Mutat. Res.* 23, 381–386.
- Cerdeira, A.L., Gazziero, D.L.P., Duke, S.O., Matallo, M.B., Spadotto, C.A., 2007. Review of potential environmental impacts of transgenic glyphosate-resistant soybean in Brazil. *J. Environ. Sci. Health B* 42, 539–549.
- Chikoye, D., Ellis-Jones, J., Riches, C., Kanyomeka, L., 2007. Weed Management in Africa: Experiences, Challenges and Opportunities. 16 International Plant Congresspp. 652–653.
- Corsello, S.M., Ubertaini, G., Altomare, M., Lovicu, R.M., Migneco, M.G., Rota, C.A., 2003. Giant prolactinomas in men: efficacy of cabergoline treatment. *Clin. Endocrinol.* 58, 662–670.
- Dallegre, E., Mantese, F.D., Oliveira, R.T., Andrade, A.J., Dalsenter, P.R., Langeloh, A., 2007. Glyphosat: pre-and postnatal toxicity of the commercial glyphosate formulation in Wistar rats. *Arch. Toxicol.* 81, 665–673.
- De Liz, O., Cavalli, V.L., Cattani, D., Heinz Rieg, C.E., Pierozan, P., Zanatta, L., 2013. Roundup disrupted male reproductive functions by triggering calcium-mediated cell death in rat testis and sertoli cells. *Free Radic. Biol. Med.* 65, 335–346.
- Dhinsa, N.K., Watson, P., Brooks, P.N., 2007. Glyphosate Technical: Dietary Two Generation Reproduction Study in the Rat. 2060/0013 BVL-2309418, ASB2012-11494. Glyphosate Technical: Dietary Two Generation Reproduction Study in the Rat. 2060/0013 BVL-2309418, ASB2012-11494.
- EFSA, 2014. The 2011 European Union report on pesticide residues in food. *EFSA J.* 12, 36–94.
- EFSA, 2016. Assessment of the Toxicological Properties of Glyphosate by the Pesticides Peer Review. ECHA RAC-39 Meeting. www.efsa.europa.eu.
- El-Tohamy, M.M., El-Nattat, W.S., 2010. Effect of antioxidant on lead-Induced oxidative damage and reproductive dysfunction in male rabbits. *J. Am. Sci.* 6 (11), 613–622.
- Garthwaite, D.G., 2010. Pesticide Usage Survey Report 235: Arable Crops In The United Kingdom 2010. Food and Environment Research Agency, DEFRA, UK.
- German Federal Institute for Occupational Safety and Health, 2016. CLH Report- Proposal for Harmonised Classification and Labelling. Based on Regulation (EC) No 1272/2008.
- Gianessi, L., Williams, A., 2011. Overlooking the obvious: the opportunity for herbicides in africa. *Outlooks Pest Manage.* 22, 211–215.
- Giri, S., Prasad, S.B., Giri, A., Sharma, G.D., 2002. Genotoxic effect of malathion: an organophosphorus insecticide, using three mammalian bioassays in vivo. *Mutat. Res.* 514, 223–231.
- Hassan, A.M., Barakat, A.H., 2008. Assessment of oxidative stress induced by nickel chloride and antioxidant effects of basil (*Ocimum basilicum* L) and thyme (*Thymus vulgaris* L). *J. Genet. Eng. Biotech.* 6 (2), 29–38.
- Hondo, E., Kurohmaru, M., Sakai, S., Ogawa, K., Hayashi, Y., 1995. Prolactin receptor expression in rat spermatogenic cells. *Biol. Reprod.* 52, 1284–1290.
- Horseman, N.D., Gregerson, K.A., 2006. Prolactin. In: DeGroot, L.J., Jameson, J.L. (Eds.), *Endocrinology*. Elsevier Saunders, Philadelphia, pp. 309–321.
- Jasper, R., Locatelli, G.O., Pilati, C., Locatelli, C., 2012. Evaluation of biochemical, hematological and oxidative parameters in mice exposed to the herbicide glyphosate-Roundup. *Interdiscip. Toxicol.* 5 (3), 133–140.
- Johansson, L.H., Borg, L.A.H., 1979. Asphotomertic method for determination of catalase activity in small tissue samples. *Anal. Biochem.* 174, 331–336.
- Kamijima, M., Hibbi, H., Gotoh, M., Taki, K., Saito, I., Wang, H., 2004. A survey of semen indices in insecticide sprayers. *J. Occup. Health* 46, 109–118.
- Larini, L., 1999. Toxicologia Dos Praguicidas. In: Manole (Ed.), pp. 178–179.
- Møller, H., 2001. Trends in incidence of testicular cancer and prostate cancer in Denmark. *Hum. Reprod.* 16, 1007–10011.
- Mesnage, R., Arno, M., Costanzo, M., Malatesta, M., Seralini, G.E., Antoniou, M.N., 2015. Transcriptome profile analysis reflects rat liver and kidney damage following chronic ultra-low dose Roundup exposure. *Environ. Health* 14, 70–75.
- Mesnage, R., Renney, G., Seralini, G.E., Ward, M., Antoniou, M.N., 2017. Multiomics reveal non-alcoholic fatty liver disease in rats following chronic exposure to an ultra-low dose of Roundup herbicide. *Sci. Rep.* 7, 39328. <http://dx.doi.org/10.1038/srep39328>.
- Monsanto International and Monsanto Europe, 2010. The agronomic benefits of glyphosate in Europe- benefits of glyphosate per market use. REVIEW 1–82.
- Mylchreest, E., Sar, M., Wallace, D., Foster, P., 2002. Fetal testosterone insufficiency and abnormal proliferation of Leydig cells and gonocytes in rats. *Reprod. Toxicol.* 16, 19–28.
- Narayana, K., D'Souza, U.J., Seetharama Rao, K.P., 2002. Ribavirin-induced sperm shape abnormalities in Wistar rat. *Mutat. Res.* 513, 193–196.
- Nithya, R., Elango, V., 2015. Pesticide effect in male hormones and antioxidant status in male albino rats. *J. Acad. Ind. Res.* 4 (4), 140–143.
- Okhawa, H., Ohishi, N., Yagi, K., 1979. Assay for lipid peroxides in animal tissues by thiobarbituric acid reaction. *Anal. Biochem.* 95, 351–358.
- Romano, R.M., Romano, M.A., Bernardi, M.M., Furtado, P.V., Oliveira, C.A., 2010. Prepubertal exposure to commercial formulation of the herbicide glyphosate alters testosterone levels and testicular morphology. *Arch. Toxicol.* 84, 309–317.
- Rotruck, J.T., Pope, A.L., Ganther, H.E., Swanson, A.B., Hafeman, D.G., Hoekstra, W.G., 1973. Selenium: biochemical role as a component of glutathione peroxidase. *Science* 179, 588–590.
- Schroeder, R.E., Hogan, G.K., 1981. Three Generation Reproduction Study in Rats with Glyphosate. 77- (BDN 77-417) BVL-1345029, TOX9552385.
- Song, L., Wang, Y.B., Sun, H., Yuan, C., Hong, X., Qu, J.H., 2008. Effects of fenvalerate and cypermethrin on rat sperm motility patterns in vitro as measured by computer-assisted sperm analysis. *J. Toxicol. Environ. Health A* 71, 325–332.
- Soon, Y.Y., Tan, B.K.H., 2002. Evaluation of the hypoglycemic and antioxidant activities of *Morinda officinalis* in streptozotocin-induced diabetic rats. *Singapore Med. J.* 43, 77–85.
- Steinmann, H.H., Dickeduisberg, M., Theuvsen, L., 2012. Uses and benefits of glyphosate

- in German arable farming. *Crop Prot.* 42, 164–169.
- Takahashi, K., 1997. HR-001: A Two-generation Reproduction Study in Rats IET 96-0031 GLP: BVL-2309425, ASB2012-11495 .
- Temple, W.A., Smith, N.A., 1992. Glyphosate herbicide poisoning experience in NewZealand. *Med. J.* 105 (933), 173–174.
- Thakur, M., Gupta, H., Singh, D., 2014. Histopathological and ultra-structural effects of nanoparticles on rat testis following 90 days (Chronic study) of repeated oral administration. *J. Nanobiotechnol.* 12, 42.
- Williams, G.M., Kroes, R., Munro, I.C., 2000. Safety evaluation and risk assessment of the herbicide Roundup and its active ingredient, glyphosate, for humans. *Regul. Toxicol. Pharmacol.* 31, 117–165.
- Williams, A.L., Watson, R.E., Desesso, J.M., 2012. Developmental and reproductive outcomes in humans and animals after glyphosate exposure: a critical analysis. *J. Toxicol. Environ. Health B Crit. Rev.* 15, 39–96.
- Xia, Y., Bian, Q., Xu, L., Cheng, S., Song, L., Liu, J., 2004. Genotoxic effects on human spermatozoa among pesticide factory workers exposed to fenvalerate. *Toxicology* 203, 49–60.
- Zitzmann, M., 2008. Effects of testosterone replacement and its pharmacogenetics on physical performance and metabolism. *Asian J. Androl.* 10 (3), 364–372.